ZIDOVEX 50 mg/5ml
P1522372

AUROBINDO

SUMMARY OF PRODUCT CHARACTERISTICS

ZIDOVEX 50 mg/5ml

Zidovudine Oral Solution 50 mg/5 mL

Rx Only

NAME OF THE FINISHED PHARMACEUTICAL PRODUCT
Zidovudine Oral Solution 50 mg/5 ml

QUALITATIVE AND QUANTITATIVE COMPOSITION
Each 5 ml of oral solution contains Zidovudine USP 50 mg

Excipients: Sucrose, glycerin, citric acid, sodium benzoate, strawberry flavor and purified water.

PHARMACEUTICAL FORM: Colourless to pale yellow strawberry flavoured syrup
filled in 300 ml white round HDPE container.

THERAPEUTIC INDICATIONS
Zidovudine is indicated in combination with other anti-retroviral agents for the treatment of Human Immunodeficiency Virus (HIV) infection in adults and children.

Zidovudine is indicated for use in HIV-positive women and the newborn infants as it has been shown to reduce the rate of maternal-foetal transmission of HIV (see Pregnancy and Lactation)

ROUTE OF ADMINISTRATION
By Oral Administration

RECOMMENDED DOSE
Zidovudine therapy should be initiated by a physician experienced in the management of HIV infection.

Adults and adolescents over 12 years of age
The recommended dose of Zidovudine in combination with other anti-retroviral agents is 500 or 600 mg/day in two or three divided doses. Dosages ≥ 1000 mg in divided doses have been used in earlier clinical trials. The effectiveness of dosages lower than 1000 mg/day in the treatment or prevention of HIV-associated neurological dysfunction is unknown.

Children (3 months to 12 years)
The recommended dose of Zidovudine is 360 to 480 mg/m2 per day, in 3 or 4 divided doses in combination with other anti-retroviral agents. For the treatment or prevention of HIV-associated neurological dysfunction, the effectiveness of dosage less than 720 mg/m2 per day (180 mg/m2 every six hours) is unknown. The maximum dosage should not exceed 200 mg every 6 hours.

Children (less than 3 months)
The limited data available are insufficient to propose specific dosage recommendations (see below maternal foetal transmission and Pharmacokinetics).

Elderly
Zidovudine pharmacokinetics has not been studied in patients over 65 years of age and no specific data are available. However, since special care is advised in this age group due to age-associated changes such as the decrease in renal function and alterations in haematological parameters, appropriate monitoring of patients before and during use of Zidovudine is advised.

Renal impairment
In patients with severe renal impairment daily dosages of 300 to 400 mg should be appropriate. Haematological parameters and clinical response may influence the need for subsequent dosage adjustment. Haemodialysis and peritoneal dialysis have no significant effect on Zidovudine elimination whereas elimination of the glucuronide metabolite is increased. For patients with end-stage renal disease maintained on haemodialysis or peritoneal dialysis, the recommended dose is 100 mg every 6 to 8 hours (see Pharmacokinetics).

Hepatic impairment
Data in patients with cirrhosis suggest that accumulation of zidovudine may occur in patients with hepatic impairment because of decreased glucuronidation. Dosage adjustments may be necessary, but as there is only limited data available precise recommendations cannot be made. If monitoring of plasma zidovudine levels is not feasible, physicians will need to monitor for signs of intolerance and adjust the dose and/or increase the interval between doses as appropriate.

Dosage adjustments in patients with haematological adverse reactions
Dosage reduction or interruption of Zidovudine therapy may be necessary in patients whose haemoglobin level falls to between 7.5 g/dl (4.65 mmol/l) and 9 g/dl (5.59 mmol/l) or whose neutrophil count falls to between 0.75 x 109/l and 1.0 x 109/l (see Contraindications, Warnings and Precautions).

Dosage in the prevention of maternal-foetal transmission
The following Zidovudine dosage regimens have been shown to be effective (see Pregnancy and Lactation).

-ACTG 076 study: The recommended dose of Zidovudine for pregnant women (over 14 weeks of gestation) is 500 mg/day orally (100 mg five times daily) until the beginning of labour. During labour and delivery Zidovudine should be administered intravenously at 2 mg/kg bodyweight given over 1 hour, followed by a continuous intravenous infusion at 1 mg/kg/h until the umbilical cord is clamped.

The newborn infants should be given Zidovudine 2 mg/kg bodyweight of oral solution every 6 hours starting within 12 hours after birth, and continuing until 6 weeks old. Infants unable to receive oral dosing should be given Zidovudine infusion intravenously at 1.5 mg/kg bodyweight infused over 30 minutes every 6 hours.

Thailand-centers for Disease Control (CDC) study. The recommended dose of Zidovudine for pregnant women from week 36 of gestation is 300 mg Zidovudine twice daily orally until onset of labour, and 300 mg Zidovudine orally every three hours from

Zidovudine therapy should be initiated by a physician experienced in the management of HIV infection.

Adults and adolescents over 12 years of age
The recommended dose of Zidovudine in combination with other anti-retroviral agents is 500 or 600 mg/day in two or three divided doses. Dosages ≥ 1000 mg in divided doses have been used in earlier clinical trials. The effectiveness of dosages lower than 1000 mg/day in the treatment or prevention of HIV-associated neurological dysfunction is unknown.

Children (3 months to 12 years)
The recommended dose of Zidovudine is 360 to 480 mg/m2 per day, in 3 or 4 divided doses in combination with other anti-retroviral agents. For the treatment or prevention of HIV-associated neurological dysfunction, the effectiveness of dosage less than 720 mg/m2 per day (180 mg/m2 every six hours) is unknown. The maximum dosage should not exceed 200 mg every 6 hours.

Children (less than 3 months)
The limited data available are insufficient to propose specific dosage recommendations (see below maternal foetal transmission and Pharmacokinetics).

Elderly
Zidovudine pharmacokinetics has not been studied in patients over 65 years of age and no specific data are available. However, since special care is advised in this age group due to age-associated changes such as the decrease in renal function and alterations in haematological parameters, appropriate monitoring of patients before and during use of Zidovudine is advised.

Renal impairment
In patients with severe renal impairment daily dosages of 300 to 400 mg should be appropriate. Haematological parameters and clinical response may influence the need for subsequent dosage adjustment. Haemodialysis and peritoneal dialysis have no significant effect on Zidovudine elimination whereas elimination of the glucuronide metabolite is increased. For patients with end-stage renal disease maintained on haemodialysis or peritoneal dialysis, the recommended dose is 100 mg every 6 to 8 hours (see Pharmacokinetics).

Hepatic impairment
Data in patients with cirrhosis suggest that accumulation of zidovudine may occur in patients with hepatic impairment because of decreased glucuronidation. Dosage adjustments may be necessary, but as there is only limited data available precise recommendations cannot be made. If monitoring of plasma zidovudine levels is not feasible, physicians will need to monitor for signs of intolerance and adjust the dose and/or increase the interval between doses as appropriate.

Dosage adjustments in patients with haematological adverse reactions
Dosage reduction or interruption of Zidovudine therapy may be necessary in patients whose haemoglobin level falls to between 7.5 g/dl (4.65 mmol/l) and 9 g/dl (5.59 mmol/l) or whose neutrophil count falls to between 0.75 x 109/l and 1.0 x 109/l (see Contraindications, Warnings and Precautions).

Dosage in the prevention of maternal-foetal transmission
The following Zidovudine dosage regimens have been shown to be effective (see Pregnancy and Lactation).

-ACTG 076 study: The recommended dose of Zidovudine for pregnant women (over 14 weeks of gestation) is 500 mg/day orally (100 mg five times daily) until the beginning of labour. During labour and delivery Zidovudine should be administered intravenously at 2 mg/kg bodyweight given over 1 hour, followed by a continuous intravenous infusion at 1 mg/kg/h until the umbilical cord is clamped.

The newborn infants should be given Zidovudine 2 mg/kg bodyweight of oral solution every 6 hours starting within 12 hours after birth, and continuing until 6 weeks old. Infants unable to receive oral dosing should be given Zidovudine infusion intravenously at 1.5 mg/kg bodyweight infused over 30 minutes every 6 hours.

Contraindications
Zidovudine is contra-indicated in patients known to be hypersensitive to zidovudine, or to any of the components of the formulations.

Warnings and Precautions
Patients should be cautioned about the concomitant use of self-administered medications (see Interactions).

Patients should be advised that therapy has not been proven to prevent the transmission of HIV to others through sexual contact or blood contamination. Appropriate precautions should continue to be taken.

Zidovudine is not a cure for HIV infection and patients remain at risk of developing illnesses which are associated with immune suppression, including opportunistic infections and neoplasms. Whilst it has been shown to reduce the risks of opportunistic infections, data on the development of neoplasms, including lymphomas, are limited. The available data on patients treated for advanced HIV disease indicate that the risk of lymphoma development is consistent with that observed in untreated patients. In patients with early HIV disease on long-term treatment the risk of lymphoma development is unknown.

Pregnant women considering the use of Zidovudine during pregnancy for prevention of HIV transmission to their infants should be advised that transmission may still occur in some cases despite therapy.

Haematological adverse reactions: Anaemia (usually not observed before 6 weeks of Zidovudine therapy but occasionally occurring earlier), neutropenia (usually not observed before 4 weeks therapy but sometimes occurring earlier) and leucopenia (usually secondary to neutropenia) can be expected to occur in patients with advanced symptomatic HIV disease receiving Zidovudine. These occurred more frequently at higher dosage (1200-1500 mg/day) and in patients with poor bone marrow reserve prior to treatment, particularly with advanced HIV disease.

Haematological parameters should be carefully monitored. For patients with advanced symptomatic HIV disease it is generally recommended that blood tests are performed at least every 2 weeks for the first 3 months of therapy and at least monthly thereafter. In patients with early HIV disease (where bone marrow reserve is generally good), haematological adverse reactions are infrequent. Depending on the overall condition of the patient, blood tests may be performed less often, for example every 1 to 3 months.

If the haemoglobin level falls to between 7.5 g/dl (4.65 mmol/l) and 9 g/dl (5.59 mmol/l) or the neutrophil count falls to between 0.75 x 109/l and 1.0 x 109/l, the daily dosage may be reduced until there is evidence of marrow recovery; alternatively, recovery may be enhanced by brief (2-4 weeks) interruption of Zidovudine therapy. Marrow recovery is usually observed within 2 weeks after which time Zidovudine therapy at a reduced dosage may be reinstituted. In patients with significant anaemia, dosage adjustments do not necessarily eliminate the need for transfusions (see Contraindications).

Lactic acidosis/severe hepatomegaly with steatosis: Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of antiretroviral nucleoside analogues either alone or in combination, including zidovudine. A majority of these cases have been in women.

Clinical features which may be indicative of the development of lactic acidosis include generalized weakness, anorexia and sudden unexplained weight loss, gastrointestinal symptoms and respiratory symptoms (dyspnoea and tachypnoea).

Caution should be exercised when administering Zidovudine to any patient, and particularly to those with known risk factors for liver disease. Treatment with Zidovudine should be suspended in any patient who develops clinical or laboratory findings suggestive of lactic acidosis or hepatotoxicity (which may include hepatomegaly and steatosis even in the absence of marked transaminase elevations).

Fat redistribution: Redistribution / accumulation of body fat, including central obesity, dorsocervical fat enlargement (buffalo hump), peripheral wasting, facial wasting and breast enlargement, elevated serum lipid and blood glucose levels have been observed either separately or together in some patients receiving combination anti retroviral therapy (see Adverse Reactions).

Whilst all members of the PI and NRTI classes of medicinal products have been associated with one or more of these specific adverse events, linked to a general syndrome commonly referred to as lipodystrophy, data indicate that there are differences in the risk between individual members of the respective therapeutic classes.

In addition, the lipodystrophy syndrome has a multi-factorial aetiology; with for example HIV disease status, older age and duration of anti retroviral treatment all playing important, possibly synergistic roles.

The long-term consequences of these events are currently unknown. Clinical examination should include evaluation for physical signs of fat redistribution. Consideration should be given to the measurement of serum lipids and blood glucose. Lipid disorders should be managed as clinically appropriate.

Immune Reconstitution Syndrome: In HIV-infected patients with severe immune deficiency at the time of initiation of anti-retroviral therapy (ART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of ART. Relevant examples are cytomegalovirus retinitis, generalized and/or focal mycobacterial infections and Pneumocystis jiroveci (P. carinii) pneumonia. Any inflammatory symptoms must be evaluated without delay and treatment initiated when necessary.

Zidovudine should not be given to patients with abnormally low neutrophil counts (less than 0.75 x 109/l) or abnormally low haemoglobin levels (less than 7.5 g/dl or 4.65 mmol/l).

Interactions with Other Medicaments
Zidovudine is primarily eliminated by hepatic conjugation to an inactive glucuronidated metabolite. Active substances which are primarily eliminated by hepatic metabolism especially via glucuronidation may have the potential to inhibit metabolism of zidovudine. The interactions listed below should not be considered exhaustive but are representative of the classes of medicinal products where caution should be exercised.

Lamivudine: A modest increase in Cmax (28%) was observed for zidovudine when administered with lamivudine, however overall exposure (AUC) was not significantly altered. Zidovudine has no effect on the pharmacokinetics of lamivudine.

Phenytoin: Phenytoin blood levels have been reported to be low in some patients receiving Zidovudine, while in one patient a high level was noted. These observations suggest that phenytoin levels should be carefully monitored in patients receiving both medicinal products.

Probenecid: Limited data suggest that probenecid increases the mean half-life and AUC of zidovudine by decreasing glucuronidation. Renal excretion of the glucuronide (and possibly zidovudine itself) is reduced in the presence of probenecid.

Ribavirin: The nucleoside analogue ribavirin antagonizes the in vitro antiviral activity of zidovudine and so concomitant use of this active substance should be avoided.

Rifampicin: Limited data sugegests that co-administration of zidovudine and rifampicin decreases AUC of zidovudine by 48% & 34%. However the clinical significance of this is unknown.

Stavudine: Zidovudine may inhibit the intracellular phosphorylation of stavudine when the two medicinal products are used concurrently. Stavudine is therefore not recommended to be used in combination with Zidovudine.

Miscellaneous: Other active substances including but not limited to aspirin, codeine, morphine, methadone, indomethacin, ketoprofen, naproxen, oxazepam, lorazepam, cimetidine, clofibrate, dapsone and Isoprinosine may alter the metabolism of zidovudine by competitively inhibiting glucuronidation or directly inhibiting hepatic microsomal metabolism. Careful thought should be given to the possibilities of interactions before using such medicinal products, particularly for chronic therapy, in combination with Zidovudine.

Concomitant treatment, especially acute therapy, with potentially nephrotoxic or myelosuppressive medicinal products (for example systemic pentamidine, dapsone, pyrimethamine, co-trimoxazole, amphotericin, flucytosine, ganciclovir, interferon, vincristine, vinblastine and doxorubicin) may also increase the risk of adverse reactions to Zidovudine. If concomitant therapy with any of these medicinal products is necessary then extra care should be taken in monitoring renal function and haematological parameters and, if required, the dosage of one or more agents should be reduced.

Since some patients receiving Zidovudine may continue to experience opportunistic infections, concomitant use of prophylactic antimicrobial therapy may have to be considered. Such prophylaxis has included cotrimoxazole, aerosolized pentamidine, pyrimethamine and acyclovir. Limited data from clinical trials do not indicate a significantly increased risk of adverse reactions to Zidovudine with these medicinal products.

Pregnancy and Lactation
Pregnancy: The use of Zidovudine prior to the 14th week of gestation should be considered only when the potential benefit to the mother outweighs the risk to the foetus.

DETACH BEFORE DISPENSING

A/s: 210 x 420 mm ■ Black

AUROBINDO Packaging Development		Product Name	Component	Item Code	Date & Time
		ZIDOVEX 50 mg/5ml	Leaflet	P1522372	10.11.2025 & 05.50 pm
		Country	Version No.	Reason of Issue	Reviewed / Approved by
		Malaysia	02	Commercial	
Team Leader	Lovaraju	Dimensions (mm)	Colours: 01	UNIT_3	
Initiator	Subramanyam	A/s: 210 x 420 mm			
Artist	MJ Designers (Kiran)	Pharma Code: 22372			
Additional Information:					

It is unknown whether there are any long-term consequences of in utero and infant exposure to zidovudine. Based on the animal carcinogenicity/mutagenicity findings a carcinogenic risk to humans cannot be excluded. The relevance of these findings to both infected and uninfected infants exposed to zidovudine in unknown. However, pregnant women considering using Zidovudine during pregnancy should be made aware of these findings.

Lactation: Women infected with HIV do not breast feed their infants.As zidovudine and the virus pass into breast milk it is recommended that mothers taking Zidovudine do not breast feed their infants.

Side Effects
The adverse event profile appears similar for adults and children. The following events have been reported in patients treated with Zidovudine. They may also occur as part of the underlying disease process in association with other medicinal products used in the management of HIV disease. The relationship between these events and use of Zidovudine is therefore difficult to evaluate, particularly in the medically complicated situations which characterize advanced HIV disease. A reduction in dose or suspension of Zidovudine therapy may be warranted in the management of these conditions.

Blood and lymphatic system disorders
Common: Anaemia (which may require transfusion), neutropenia and leucopenia.

These occur more frequently at higher dosages (1200-1500 mg/day) and in patients with advanced HIV disease (especially when there is poor bone marrow reserve prior to treatment), and particularly in patients with CD4 cell counts less than 100/mm3. Dosage reduction or cessation of therapy may become necessary (see Warnings and Precautions). The incidence of neutropenia was also increased in those patients whose neutrophil counts, haemoglobin levels and serum vitamin B12 levels were low at the start of Zidovudine therapy.

Uncommon: Thrombocytopenia and pancytopenia (with marrow hypoplasia).
Rare: Pure red cell aplasia.
Very rare: Aplastic anaemia.

Metabolism and nutrition disorders
Common: Hyperlactataemia.
Rare: Lactic acidosis (see Warnings and Precaurions), anorexia.

Redistribution/accumulation of body fat (see Warnings and Precautions). The incidence of this event is dependent on multiple factors including the particular antiretroviral drug combination.

Psychiatric disorders
Rare: Anxiety and depression.
Nervous system disorders
Very common: Headache.
Common: Dizziness.
Rare: Insomnia, paraesthesia, somnolence, loss of mental acuity, convulsions.

Cardiac disorders
Rare: Cardiomyopathy.

Respiratory, thoracic and mediastinal disorders
Uncommon: Dyspnoea.
Rare: Cough

Gastrointestinal disorders
Very common: Nausea.
Common: Vomiting, abdominal pain, and diarrhea.
Uncommon: Flatulence.
Rare: Oral mucosa pigmentation, taste disturbance and dyspepsia, Pancreatitis.

Hepatobilary disorders
Common: Raised blood levels of liver enzymes and bilirubin.
Rare: Liver disorders such as severe hepatomegaly with steatosis.

Skin and subcutaneous tissues disorders
Uncommon: Rash and pruritus.
Rare: Nail and skin pigmentation, urticaria and sweating.

Musculoskeletal and connective tissue disorders
Common: Myalgia.
Uncommon: Myopathy.

Renal and urinary disorders
Rare: Urinary frequency.

Reproductive system and breast disorders
Rare: Gynaecomastia.

General disorders and administration site conditions
Common: Malaise.
Uncommon: Fever, generalized pain and asthenia.
Rare: Chills, chest pain and influenza-like syndrome.

The available data from both placebo-controlled and open-labeled studies indicate that the incidence of nausea and other frequently reported clinical adverse events consistently decreases over time during the first few weeks of therapy with Zidovudine.

Adverse reactions with Zidovudine for the prevention of maternal-foetal transmission
Zidovudine was well tolerated in pregnant women at the doses recommended for this indication. Clinical adverse events and laboratory test abnormalities were similar in the Zidovudine and placebo groups.Haemoglobin concentrations in infants exposed to Zidovudine for this indication were marginally lower than in infants in the placebo group, but transfusion was not required. Anaemia resolved within 6 weeks after completion of Zidovudine therapy. Other clinical adverse events and laboratory test abnormalities were similar in the Zidovudine and placebo group. The long-term consequences of in utero and infant exposure to Zidovudine are unknown.

Symptoms and Treatment of Overdose
Symptoms and signs:
No specific symptoms or signs have been identified following acute overdose with Zidovudine, apart from those listed as undesirable effect such as fatigue, headache, vomiting, and occasional reports of haematological disturbances. Following a report where patients took an unspecified quantity of Zidovudine, blood zidovudine levels were over sixteen times the normal therapeutic level, but there were no short term clinical, biochemical or haematological sequelae identified.

Treatment:
Patients should be observed closely for evidence of toxicity (see Adverse Reactions) and given the necessary supportive therapy. Haemodialysis and peritoneal dialysis appear to have a limited effect on elimination of zidovudine but enhance the elimination of the glucuronide metabolite.

Pharmacodynamics: ATC Code: J05AF01 - ZIDOVUDINE

Zidovudine is an antiviral agent which is highly active in vitro against retroviruses including the Human Immunodeficiency Virus (HIV). Zidovudine is phosphorylated in both infected and uninfected cells to the monophosphate (MP) derivative by cellular thymidine kinase. Subsequent phosphorylation of zidovudine-MP to the disphosphate (DP), and then the triphosphate (TP) derivative is catalysed by cellular thymidylate kinase and non-specific kinases respectively. Zidovudine-TP acts as an inhibitor of and substrate for the viral reverse transcriptase. The formation of further proviral DNA is blocked by incorporation of zidovudine-TP into the chain and subsequent chain termination. Competition by zidovudine-TP for HIV reverse transcriptase is approximately 100-fold greater than for cellular DNA polymerase alpha. Zidovudine has been shown to act additively or synergistically with a number of anti-HIV agents, such as lamivudine, didanosine and interferon-α, inhibiting the replication of HIV in cell culture.

Resistance to thymidine analogues (of which zidovudine is one) is well characterized and is conferred by the stepwise accumulation of up to six specific mutations in the HIV reverse transcriptase at codons 41, 67, 70, 210, 215 and 219. Viruses acquire phenotypic resistance to 41 and 215 or by the accumulation of at least four of the six mutations. These thymidine analogue mutations alone do not cause high-level cross-resistance to any of the other approved reverse transcriptase inhibitors.

Two patterns of multi-drug resistance mutations, the first characterized by mutations in the HIV reverse transcriptase at codons 62, 75, 77, 116 and 151 and the second typically involving a T69S mutation plus a 6-base pair insert at the same position, result in phenotypic resistance to zidovudine as well as to the other approved nucleoside reverse transcriptase inhibitors. Either of these two patterns of multinucleoside resistance mutations severely limits future therapeutic options.

Reduced in vitro sensitivity to zidovudine has been reported for HIV isolates from patients who have received prolonged courses of Zidovudine therapy. The available information indicates that for early HIV disease, the frequency and degree of reduction of in vitro sensitivity is notably less than for advanced disease.

The relationships between in vitro susceptibility of HIV to zidovudine and clinical response to therapy remain under investigation. In vitro susceptibility testing has not been standardized and results may therefore vary according to methodological factors.

Studies in vitro of zidovudine in combination with lamivudine indicate that zidovudine-resistant virus isolates can become zidovudine sensitive when they simultaneously acquire resistance to lamivudine. Evidence from clinical studies show that lamivudine plus zidovudine delays the emergence of zidovudine-resistant isolates in individuals with no prior anti-retroviral therapy.

Zidovudine has been widely used as a component of anti retroviral combination therapy with other anti retroviral agents of the same class (nucleoside reverse transcriptase inhibitors) or different classes (protease inhibitors, non-nucleoside reverse transcriptase inhibitors).

Post-exposure prophylaxis (PEP)
Internationally recognized guidelines (Centre for Disease Control and Prevention-June 1998), recommend that in the event of accidental exposure to HIV infected blood e.g. from a needlestick injury, a combination of Zidovudine and Lamivudine should be administered promptly (within one to two hours). In cases of higher risk of infection a protease inhibitor should be included in the regimen. It is recommended that antiretroviral prophylaxis be continued for four weeks. No controlled clinical studies have been carried out in post-exposure prophylaxis and supporting data is limited. Seroconversion may still occur despite prompt treatment with antiretroviral agents.

Pharmacokinetics
Absorption
Zidovudine is well absorbed from the gut and, at all dose levels studied; the bioavailability was 60 to 70%. From a Phase 1I study, mean steady state peak (C[ss] max) and through (C[ss]min) plasma concentrations following oral administration of zidovudine (in solution) at doses of 5 mg/kg every 4 h were 7.1 and 0.4 micromol (or 1.9 and 0.1 micrograms/ml) respectively. From a bioequivalence study, mean C[ss]max and C[ss]min levels following oral administration of zidovudine capsules every 4 h and dose normalized to 200 mg were 4.5 micromol (or 1.2 micrograms/ml) and 0.4 micromol (or 0.1 micrograms/ml) respectively.

Distribution
From studies with i.v.zidovudine, the mean terminal plasma half-life was 1.1 h, the mean total body clearance was 27.1 ml/min/kg and the apparent volume of distribution was 1.6 l/kg. In adults, the average cerebrospinal fluid/plasma zidovudine crosses the placenta and is found in amniotic fluid and foetal blood. Zidovudine has also been detected in semen and milk. Plasma protein binding is relatively low (34 to 38%) and interactions with other active substances involving binding site displacement is not anticipated.

Metabolism
The 5-glucuronide of zidovudine is the major metabolite in both plasma and urine, accounting for approximately 50 to 80% of the administered dose eliminated by renal excretion. 3-amino-3-deoxythymidine (AMT) has been identified as a metabolite of zidovudine following i.v. dosing.

Elimination
Renal clearance of zidovudine greatly exceeds creatinine clearance, indicating that significant tubular secretion takes place.

Special Patient Populations
Children
In children over the age of 5 to 6 months, the pharmacokinetic profile of zidovudine is similar to that in adults.

Zidovudine is well absorbed from the gut and, at all dose levels studied; its bioavailability was 60 to 74% with a mean of 65%. C[ss] max levels were 4.45 micromol (1.19 micrograms/ml) following a dose of 120 mg zidovudine (in solution)/ m2 body surface area and 7.7 micromol (2.06 micrograms/ml) at 180 mg/m2 body surface area.

With i.v. dosing, the mean terminal plasma half-life and total body clearance were 1.5 h and 30.9 ml/min/kg respectively. The major metabolite is the 5-glucuronide. After i.v. dosing, 29% of the dose was recovered unchanged in the urine and 45% excreted as the glucuronide. Renal clearance of zidovudine greatly exceeds creatinine clearance indicating that significant tubular secretion takes place.

The data available on the pharmacokinetics in neonates and young infants indicate that glucuronidation of zidovudine is reduced with a consequent increase in bioavailability, reduction in clearance and longer half-life in infants less than 14 days old but thereafter the pharcokinetics appear similar to those reported in adults.

Elderly
The pharmacokinetics of zidovudine has not been studied in patients over 65 years of age.

Renal Impairment
Compared to healthy subjects, patients with advanced renal failure have a 50% higher peak plasma concentration of zidovudine. Systemic exposure (measured as area under the zidovudine concentration-time curve) is increased 100%; the half-life is not significantly altered. In renal failure there is substantial accumulation of the major, glucuronide metabolite but this does not appear to cause toxicity. Haemodialtsis and peritoneal dialysis have no significant effect on zidovudine elimination whereas elimination of the glucuronide metabolite is increased (see Dosage and Administration).

Hepatic Impairment
Data in patients with cirrhosis suggest that accumulation of zidovudine may occur in patients with hepatic impairment because of decreased glucuronidation. Dosage adjustment may be necessary, but as there is only limited data available precise recommendations cannot be made (see Dosage and Administration).

Pregnancy
The pharmacokinetics of zidovudine has been investigated in a study of eight women during the last trimester of pregnancy. As pregnancy progressed, there was no evidence of accumulation of zidovudine. The pharmacokinetics of zidovudine was similar to that of non-pregnant adults. Consistent with passive transmission of zidovudine across the placenta, zidovudine concentrations in infant plasma at birth were essentially equal to those in maternal plasma at delivery.

Storage Condition
Do not store above 30 °C. Store in the original package. Keep the container tightly closed.

Shelf Life
24 months

Manufactured by:

AUROBINDO

Aurobindo Pharma Limited
Unit III Survey No.: 313,314, Bachupally Village, Bachupally Mandal, Medchal Malkajgiri District, Hyderabad, Telangana, 500090, India

Registered Office:
#2, Maitrivihar, Ameerpet,
Hyderabad-500 038, INDIA

Product Registration Holder in Malaysia:
Unimed Sdn Bhd, No. 53, Jalan Tembaga SD,
5/2B, Bandar Sri Damansara, 52200,
Kuala Lumpur, Malaysia.

Date of Revision: Oct 2025

P1522372

DETACH BEFORE DISPENSING

the following medicines (ask your doctor if you are not sure) :

- Phenytoin, sodium valproate
- Stavudine, ribavirin, ganciclovir, interferon
- Pyrimethamine, dapsone, fluconazole, flucytosine
- Pentamidine, atovaquone, rifampicin, amphotericin, clarithromycin, co-trimoxazole
- Probenecid, methadone
- Vincristine, vinblastine, doxorubicin

In patients who take vitamin supplements, herbal remedies or homeopathic medicines only a few side effects have been seen, so if you feel benefit from these, continue taking them but keep your doctor informed.

3. How to take zidovudine oral solution?

It is important to take your medicine in the way your doctor has told you to. If you forget to take a dose, don't worry. Simply take it as soon as you remember and then continue as before. If you think you may have difficulty in remembering to take your medicine at the times specified, it's a good idea to use a pocket timer or wristwatch alarm to remind you. If you take a larger dose than prescribed this is not likely to cause you any harm, but you should let your doctor know as soon as possible if this happens.

If someone else takes your medicine by mistake, tell your doctor at once.

Because your medicine controls and does not cure your condition, you will normally need to take it continuously. You should not stop treatment unless your doctor tells you to.

Remember that treatment with zidovudine oral solution does not reduce the risk of passing HIV on to others by sexual contact or blood transfer. You will still remain infectious while taking zidovudine oral solution.

The dose prescribed will vary from patient to patient and will depend on a number of factors, including the stage of infection. Your doctor will decide the most appropriate dosage for you.

If you take more zidovudine oral solution than you should:
If you take a larger dose than prescribed this is not likely to cause you any harm, but you should let your doctor know as soon as possible if this happens.

If you forget to take zidovudine oral solution :
If you forget to take a dose, don't worry. Simply take it as soon as you remember and then continue as before. If you think you may have difficulty in remembering to take your oral solution at the times specified, it's a good idea to use a pocket timer or wrist-watch alarm to remind you.

4. Possible side effects of zidovudine oral solution

Zidovudine oral solution is normally well tolerated. In patients with early HIV infection side effects are less common than in patients with AIDS. However, because of the way zidovudine oral solution works, it may sometimes cause side effects. If you notice any of these side effects, tell your doctor.

- Most commonly, production of red blood cells is reduced, resulting in anaemia. If this happens, the symptoms are tiredness and shortness of breath.
  In this case, the doctor is likely to either reduce your dose or temporarily interrupt your treatment, usually for between two and four weeks, in order to allow your blood time to recover. Less commonly, the production of a certain type of white blood cell may be reduced which can make you more prone to infections. Your doctor may want you to have a blood test from time to time to check the blood cell count. These effects are generally reversible and it is important to note that, with the lower treatment doses currently being prescribed, anaemia is now less common than when zidovudine oral solution was first introduced.
- Nausea, vomiting, changes in skin colour inside the mouth, stomach pain, heartburn, loss of appetite, diarrhoea and trapped air in the stomach. In the vast majority of cases, all these effects disappear on their own after a few weeks, a fact worth bearing in mind if you experience them. Check with your doctor if they do not go away or if they become distressing.
- Headache, dizziness, pins and needles, inability to concentrate, seizures, depression, feeling anxious and not being able to sleep.
- Breathlessness and cough.
- Nail and skin colour changes, rash (red, raised or itchy) and sweating.
- Liver disorders eg enlarged liver, fatty liver and jaundice.
- Inflammation of the pancreas which makes digestive juices and insulin.
- Muscular aches may develop.
- "Flu-like" feeling, fever, tiredness, chills, chest pain, enlarged breasts in male patients, taste changes, general aches and pains and passing urine more frequently.
- Cardiomyopathy (disease of the heart muscle). Symptoms of this may include shortness of breath, swelling of your ankles or fluid in your lungs.
- A rare but serious condition called lactic acidosis, together with an enlarged liver, can develop in some people taking zidovudine oral solution. This is where a build-up of lactic acid in the body can lead to a coma and can sometimes be life threatening. As the condition develops, you may experience stomach pain, nausea (feelings of sickness), vomiting, breathing problems or numbness. In severe cases, inflammation of the pancreas, liver or kidney failure or an increase in lactate in the body may develop. If you have liver disease or if you are an obese (very overweight) woman, you are at more risk of getting this condition which generally comes on after a few months treatment.
- Redistribution, accumulation or loss of body fat may occur in patients receiving combination antiretroviral therapy. Combination antiretroviral therapy may cause changes in body shape due to changes in fat redistribution. These may include loss of fat from legs, arms and face, increased fat in the abdomen (belly) and other internal organs, breast enlargement and fatty lumps on the back of the neck ('buffalo hump'). The cause and long-term effects of these conditions are not known at this time. Combination antiretroviral therapy may also cause raised lactic acid sugar in the blood,hyperlipaemia (increased fats in the blood) and resistance to insulin. Contact your doctor if you notice changes in body fat.

Tell your doctor or pharmacist if you notice any other side effects from your medicine which are not mentioned here.

5. Storing zidovudine oral solution
Keep your medicine in a safe place. Store below 30°C. Store the bottle in the original outer carton. Keep it out of the reach and sight of children.

Do not take any medicine after the expiry date on the pack.

REMEMBER: THIS MEDICINE IS FOR YOUR USE ONLY. Never give it to others as it may harm them even if their symptoms are the same as yours.

6. Further information
If you notice any side effect (s) with the use of this drug, please report it immediately via internet to the following e-mail address: pharmacovigilance@aurobindo.com.
If you have questions or concerns, or want more information about zidovudine oral solution, consult your doctor. This medicine was prescribed for your particular condition. Do not use zidovudine oral solution for another condition.